



IoT Security

A primer on the concepts of security in the chaotic and ever growing world of IoT.

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Regardless of whether you are a hacker or not, you must've watched Mr.Robot (if you haven't, what are you waiting for?). We would like to draw your attention to season 2's premiere episode *spoiler alert*. One of the hackers (the chaotic good) breaks into one of the bad guys' house and *actually* hacks her home! The lights, the doors, the water-works, and even the thermostat. The idea of smart "things" doesn't seem so appealing after that, does it?

Internet of Things (IoT) is the buzzword these days where "things" are being interconnected with each other and traditional devices. The things here are devices that weren't considered "digital" before. For example, the light bulb in your room can be connected to your phone or the refrigerator can send you updates as soon as the ice cubes are ready. This is just a peek into the IoT world. Everything from automobiles to vacuum cleaners are caught in this IoT wave, making them smarter and us lazier. But, with every emerging technology, comes the compromises. We dive into the plethora of methods in which IoT/smart devices weaken the security of the system infrastructure they are a part of, so you can be lazy and secure at the same time.

THE PREDICAMENTS:

IoT security is one of the most overlooked and neglected aspects when

such devices are developed. You might be wondering, well what's the need for securing these devices? Who's going to benefit from hacking a light bulb? Look at it this way. Your big luxurious house is connected to a small old shack, with a very weak lock on it because who will break into an unimportant shack. But the shack leads to the house and the results are devastating. Unsecured IoT devices are the shacks to the house that is your whole network system.

They act as easy to exploit entry points which allow attackers to move laterally through a network, increasing the attack severity and disclosure of sensitive data. Also, the number of IoT devices being employed by different industries is so huge, they are not considered "worthless entry points" anymore. According to reports, about 87% of healthcare organizations have adopted some sort of IoT device as of 2019.

Health details can be extracted from medical devices, personal information can be stolen from smart speakers, arbitrary and possibly malicious commands can be executed on automobile firmware (remember the popular Tesla hack? [<https://dgit.in/teshack>]) and the list goes on. The Checkpoint Security Research group showed how easy it is to hack an ultrasound machine because the underlying operating system on which the machine runs was Windows 2000, which reached its end-of-life phase on 1st January 2017. They used Metasploit (a penetration framework, providing information about numerous vulnerabilities and pre-built modules to exploit) modules through a python script to exfiltrate images out of the ultrasound machine and it is not something that demands extraordinary skills. That right there is the number one reason smart devices get compromised: the code that powers the device is outdated. Most of the devices being created are based on old and legacy firmware, with numerous vulnerabilities present in them. Since vendors have moved on to focus on the current versions and technologies, these vulnerabilities stay unfixed and ready to be exploited. Apart from this major cause, other factors like weak or hard coded